

Metal laminates based on waterborne epoxy varnishes - chemical interactions and effect on fatigue crack growth kinetics

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ABSTRACT

Epoxy coatings cured with the latent curing agent dicyandiamide (DICY) are well-established. During coating of waterborne epoxy varnishes onto the metal substrate, chemical interactions take place, which were evaluated in this paper for different metal substrates. Moreover, their effect on the fatigue crack growth kinetics of bonded laminates was assessed. First, different pure metal sheets were immersed in a boiling solution of DICY in distilled water. Metals with a lower isoelectric point (IEP) ($\text{pH} < 5$) revealed the formation of a higher amount of protonated amine. In contrast, for metals with a higher IEP ($\text{pH} > 8$) the formation of imine was preferred. Furthermore, epoxy coated metal sheets were prepared by cryo ultra-low-angle microtomy (cryo-ULAM) and the reaction products were elucidated by X-ray photoelectron spectroscopy (XPS). Close to the metal/epoxy interface, reaction products were ascertained, which corroborated the results of the immersion test. Finally, the effect of the IEP of the substrate on the crack growth kinetics of bonded laminates was evaluated by cyclic fatigue experiments on double cantilever beam specimens. The higher the IEP of the metal substrate, the better was the crack growth resistance in the threshold and stable crack growth regime. Overall, this study clearly confirmed that chemical interactions are taking place during manufacturing of metal/waterborne epoxy laminates, which have a significant impact on the adherence and crack growth resistance.

1. Introduction

Epoxy based varnishes are used for protective coatings, as adhesives or as encapsulation materials [1–6]. For crosslinking of the epoxy resin, which is commonly based on Diglycidyl ether of bisphenol-A (DGEBA), amine curing agents are added. Due to its latent nature, dicyandiamide (DICY) is state-of-the-art for high temperature crosslinking adhesives and coatings [7–10]. As confirmed for solvent-borne epoxies, aliphatic amine hardeners are interacting with metal oxides associated with the formation of an interphase [7,8,11–15]. The chemical structure of the interphase with a thickness in the micrometre range [12] is different to the bulk of epoxy thermosets.

X-ray photoelectron spectroscopy (XPS) is a versatile tool for the elucidation of interfacial interactions of amine curing agents with metal

surfaces [13–15]. Interactions of DICY and selected metal surfaces were confirmed by Bengu et al. [7,8] and in a previous associated paper [8]. The formation of the interphase depends on the substrate [7,8,11,14–16]. Many different factors including morphological parameters such as the oxide/hydroxide thickness, roughness and crystallography have an impact on the interphase formation. Furthermore, physiochemical parameters such as the oxide/hydroxide balance [14,17–19], the surface treatment [16], the nature of the amine [20] and the isoelectric point (IEP) are affecting the network structure. However, in literature only few studies exist which investigate the influence of the IEP on the interphase formation and the adherence under service relevant conditions [21]. While Sn, Al, and Zn were assessed in the associated paper [8], the main objective of this study is to do a more comprehensive investigation considering Sn, Ti, Fe, Al, Zn and Mg and

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to establish potential correlations between the IEP of the metals and the reaction products.

Waterborne epoxy varnishes are aqueous emulsions, in which the DICY hardener is mainly dissolved in the water phase [8,10,22–24]. In water environment, metal surfaces get electrically charged. The IEP is equivalent to the pH value of an aqueous solution in contact with the metal surface resulting in a net charge of zero [25,26]. To evaluate metal/DICY interactions, the following methodological approach was used: First, metal samples were immersed in a H_2O + DICY solution at the boiling temperature (see Fig. 1a). Second, metal substrates were coated with waterborne epoxy varnishes (see Fig. 1b). In the cured state, different depths were investigated by tapers produced cryo-ultra-low-angle microtomy (ULAM), which were characterized by XPS. Third, laminates and double cantilever beam (DCB) specimens were prepared and examined with respect to the fatigue crack growth behavior at 23 °C and 50 % relative humidity (rh). Additionally, the fractured surfaces were examined by XPS (see Fig. 1c).

2. Methodology and approach

2.1. Immersion and coating of metal substrates

Pure metal sheets based on Sn, Ti, Fe, Al, Zn and Mg (thickness: 0.5 to 1.0 mm) were purchased from Polymet (Germany). Due to the fact that no specific Young's modulus (E) and yield strength (σ_y) data were given in the data sheets, the following representative value ranges were taken from literature data: Sn: $E = 47\text{--}70$ GPa, $\sigma_y = 10\text{--}20$ MPa; Ti: $E = 116$ GPa, $\sigma_y = 100\text{--}225$ MPa; Fe: $E = 204\text{--}212$ GPa, $\sigma_y = 100\text{--}250$ MPa; Al: $E = 69\text{--}70$ GPa, $\sigma_y = 35$ MPa; Zn: $E = 83\text{--}110$ GPa, $\sigma_y = 70\text{--}110$ MPa; Mg: $E = 42\text{--}45$ GPa, $\sigma_y = 200\text{--}270$ MPa [27–32]. Prior to XPS analysis, the metal substrates were cleaned alternately with tetrahydrofuran and isopropanol in an ultrasonic bath for 30 min for two consecutive times.

For immersion testing, the metal sheets were cut into smaller pieces

of approximately $5 \times 5 \text{ mm}^2$. A solution of DICY was prepared by diluting pure DICY powder (Sigma-Aldrich, Austria) in water. The concentration of the hardener was 3 wt%. The removed metal samples were rinsed with water and then transferred to the XPS chamber for analysis.

For investigation on coated metal level, a waterborne epoxy model varnish was used. Details of the formulation and processing are given in [22]. The epoxy equivalent weight (EEW) was ranging from 860 to 930 g/mol. The amine H equivalent weight of DICY is 21 based on the 4 reactive sites [33]. Based on this assumption, the DICY content in parts per hundred (PHR) was calculated. An over-stoichiometric ratio of 1.1 to 1 (DICY to epoxy) was chosen. As determined in previous work, a glass transition peak temperature of 91 °C was deduced by Dynamic Mechanical Analysis (DMA) from the loss factor curve [22]. The varnish was applied onto the metal sheets with a spiral bar and subsequently pre-cured. The average coating thickness in the dried state was 5 μm . Afterwards, the coatings were fully cured at 200 °C for 2 h under N_2 atmosphere. The coated metal sheets were cut into smaller pieces of $4 \times 12 \text{ mm}^2$ and glued onto a customized aluminum sample holder. Ultra-low-angle tapers were prepared with a Leica Ultracut UCT ultramicrotome using a standard glass knife. By tilting the sample holder by an angle of $\leq 3^\circ$, a wedge shaped section was obtained, revealing the underlying coating and the substrate. To minimize smearing, the coated metal sheet was cooled with liquid nitrogen to 0 °C (cryo-ULAM).

2.2. Preparation and testing of laminates

Double cantilever beam (DCB) laminate specimens were prepared using pure metal substrates and a waterborne epoxy varnish with an EEW of approximately 500 g/mol since a lower mass epoxy allows an easy preparation of the simple epoxy varnish. The stoichiometric ratio of DICY to epoxy was 1.1 to 1. Dynamic mechanical analysis on the epoxy films in tension mode revealed a Young's modulus E of 2500 MPa at

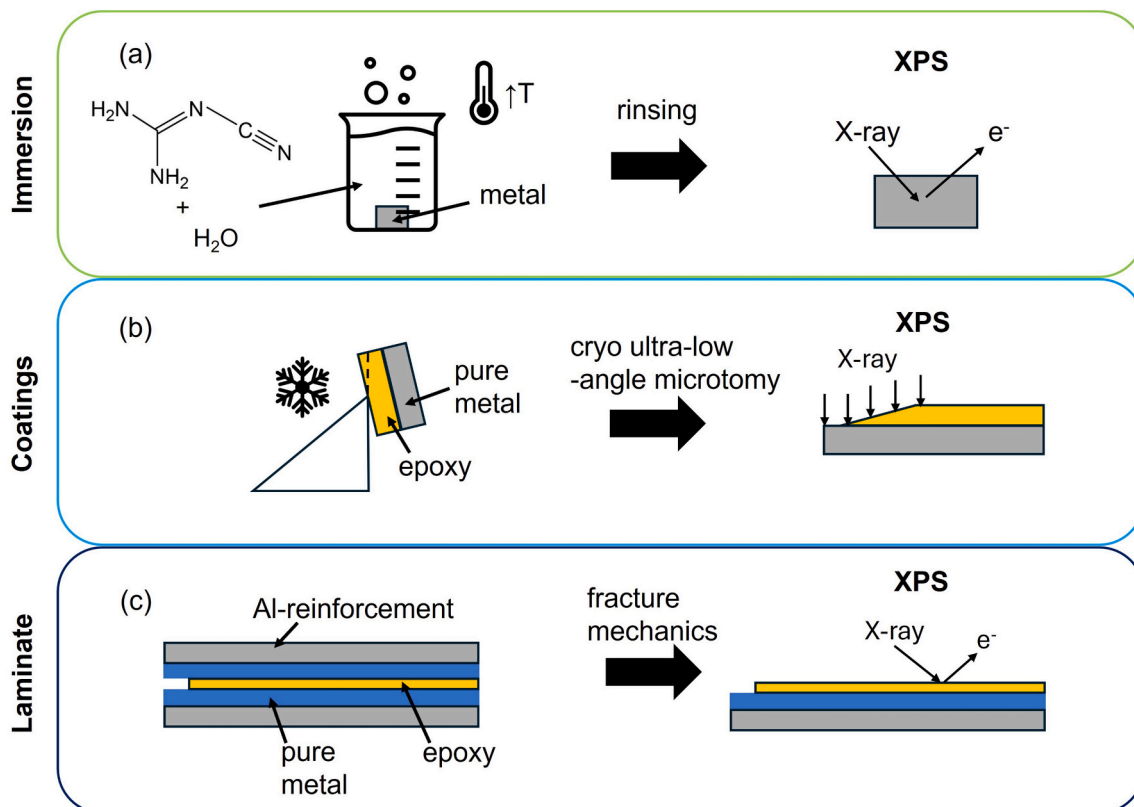


Fig. 1. Schematic sketch of (a) the H_2O + DICY mixture immersion test, (b) assessment of coatings by cryo-ULAM and XPS and (c) fracture mechanics fatigue testing of laminates.

23 °C and a glass transition temperature deduced from the loss factor peak of 102 °C [22]. The dimension of the DCB specimens were $150 \times 25 \text{ mm}^2$. Hence, the metal sheets were cut on a CNC milling machine (EMCO E600, EMCO, Austria). Subsequently, the metal sheets were sanded with 180-grit sandpaper. The substrates were coated in a multi-step process. First, the metal sheets were cleaned with isopropanol. Then, the varnish in the uncured A-stage was applied using a spiral bar coater giving a wet film thickness of 20 μm . To dry and partly cure the coating to the so-called B-stage, the coated metal sheets were placed in an oven at a peak metal temperature of 210 °C for 25 s. Coated metal sheets in B-stage have a shelf lifetime of up to 6 months. Nevertheless, the coated metal sheets were immediately used to manufacture the laminated specimen. To obtain DCB specimen, a thin silver leaf ($<0.25 \mu\text{m}$) was inserted in the middle to ensure an artificial crack of well-defined size. Laminates consisting of 2 metal plates were stacked. The epoxy plies were fully cured to the C-stage in a heated hydraulic press at a temperature of 200 °C for 1 h under an applied pressure of around 0.5 MPa. Since the pure metal sheets were rather thin and unsuitable for direct fatigue testing, an aluminum sheet with a thickness of 3 mm was applied on each side as reinforcement using a two-component epoxy adhesive (DP490, 3 M, USA). To introduce the delamination force, load blocks were mounted with a DP490 adhesive. A schematic drawing of the symmetric DCB specimen structure is depicted in Fig. 2a.

Fatigue tests were conducted on an electrodynamic test machine (Instron ElectroPuls E1000, Instron, USA) equipped with a 2 kN load cell (Dynacell, Instron, USA). The tests were carried out at 23 °C and 50 % rh with a maximum displacement amplitude of 1.5 mm. The R-ratio, which defines the ratio of minimum to maximum displacement, was set to 0.1, resulting in a displacement amplitude of 0.675 mm. A photograph of the test setup is shown in Fig. 2b.

To compare the delamination resistance of the DCB specimen, fatigue crack growth curves were analyzed (see Fig. 2c). The crack propagation rate (da/dN) was plotted as a function of the strain energy release rate (SERR or G). A shift to higher SERR and lower da/dN values indicates enhanced delamination resistance.

fatigue tests allow for measuring the fatigue threshold value at which the crack propagation is negligible (region I, Fig. 2c). To monitor and estimate the crack length during testing, an elastic foundation model was employed [34,35]. Data reduction and evaluation was carried out according to [34,35].

2.3. X-ray photoelectron spectroscopy (XPS)

Two systems were used for XPS analysis. The immersed metal samples and most of the ULAM tapers were analyzed with a Theta Probe system (ThermoFisher, UK). The ULAM cut of the Fe coated sample and the fractured double cantilever beams were characterized with a Nexsa G2 system (ThermoFisher, UK). For both XPS devices, the Advantage software package, as provided by ThermoFisher, was used for acquisition and evaluation. A monochromatic Al K_{α} X-ray source (1486.6 eV) was employed. Charge compensation was carried out with a dual flood gun. A maximal X-ray spot size of 400 μm was selected for the metal substrates in the reference state and after immersion in the $\text{H}_2\text{O} + \text{DICY}$ mixture. For the ULAM tapers, the spot size was 150 μm for the Theta probe system and 50 μm for the Nexsa system. The step size was set in both cases to 50 μm . For charge correction, the spectra were shifted according to the C-C/C-H peak at 285.0 eV. Background correction was performed with the function smart background of the Advantage evaluation software. The elemental composition was acquired by using the Scofield sensitivity factors provided by the manufacturer. For fitting of the XPS high resolution spectra, a product of a Lorentzian and Gaussian shaped peak was utilized.

3. Results and discussion

3.1. Structural changes due to immersion of metals in aqueous DICY solutions

As described in a previous paper [8], immersion of Al, Sn and Zn metals in the $\text{H}_2\text{O} + \text{DICY}$ solution at boiling temperatures resulted in

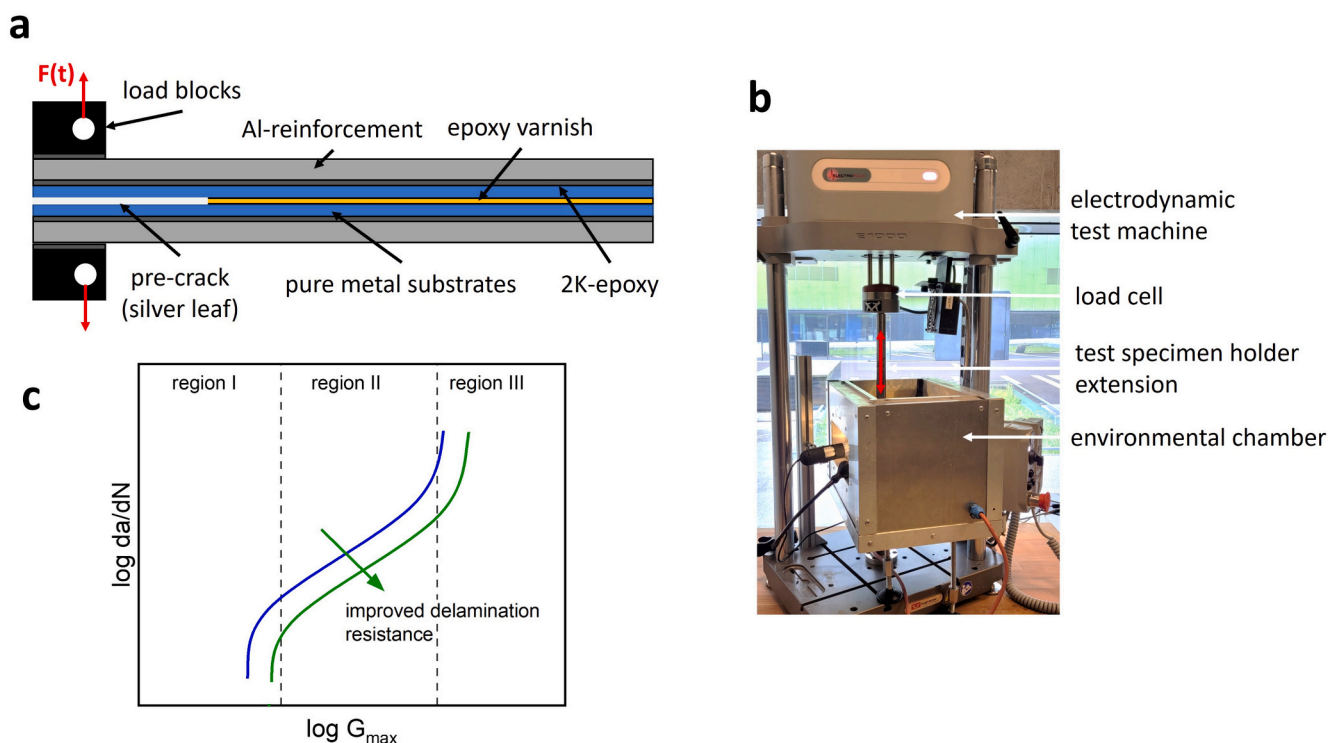


Fig. 2. (a) Sketch of a DCB specimen, (b) the test setup for fatigue crack growth experiments and (c) crack propagation rate plotted (da/dN) as a function of maximum strain energy release rate (SERR or G).

significant changes of the chemical structure on the surface of the passivation layer. As a reference the specimens were also tested in pure water. In this paper, more comprehensive immersion tests were carried out considering also Ti, Fe and Mg metal substrates. The IEP of oxides and hydroxides of the investigated metals are summarized in Table 1. The values were mainly taken from reference [25]. According to literature, Sn based oxides and hydroxides are acidic (IEP: 4.0–5.2), while Mg oxides and hydroxides are classified as basic or alkaline (IEP: 10.0–12.5).

The high resolution XPS spectra of the reference and immersed metal surfaces are depicted in Fig. 3. In the reference Sn3d spectrum, a metallic and oxidic photoelectric peak were discernible (see Fig. 3a). After immersion in water or H₂O + DICY mixture, the metallic component disappeared. Hence, the oxide layer got thicker than the information depth of XPS (~10 nm).

No significant changes were detectable for the Ti2p peak (see Fig. 3b). Before and after treatment, the associated photoelectric peak of metallic Ti was not visible in the spectra. The Ti oxide passivation layer was already thicker than 10 nm in the non-immersed reference state.

The Fe sample revealed satellite peaks at approximately 719.5 eV and 733 eV (see Fig. 3c), which confirmed that the investigated Fe oxide is a Fe³⁺ compound based on Fe₂O₃ or FeOOH [36,37]. After immersion, the Fe³⁺ characteristics were also observed. Furthermore, the metallic Fe photoelectric peak disappeared indicating again the growth of the oxide layer to a thickness of >10 nm due to immersion in boiling water or H₂O + DICY mixture.

Immersion had also an effect on the Al sample (see Fig. 3d). In agreement with the other metals, the oxide got thicker. However, Al revealed a peak shift and narrowing of the oxidic Al2p peak width, most likely due to hydroxylation of the oxide layer.

Interestingly, a different reaction mechanism was observable for Zn immersed in boiling water or H₂O + DICY mixture (see Fig. 3e). The ZnLMM oxidic peak changed significantly. As described in [8], the modified Auger parameter (MAP) was 2009.9 eV (ZnO [38]) in the reference state and after immersion in boiling water. However, immersion boiling in the H₂O + DICY mixture led to a shift to 2009.1 eV indicating the formation of Zn(OH)₂ [38]. Independent of the immersion conditions, the ZnO and Zn(OH)₂ layer was growing significantly, achieving a passivation layer thickness exceeding 10 nm.

For the Mg sample, the MgKLL Auger peak was evaluated (see Fig. 3f). No metallic Mg peak was detectable for the reference state. Hence, the initial Mg oxide layer thickness already exceeded 10 nm. After immersion in water or H₂O + DICY mixture, the MgKLL spectra were shifted to slightly lower binding energy values. The MAP stayed nearly constant (2483.9 eV in the reference state, 2484.0 eV after immersion in water or H₂O + DICY). Presumably, the passivation layer on the Mg sample was primarily based on Mg(OH)₂.

To assess potential interactions of DICY with the metal surfaces, N1s XPS spectra were recorded and evaluated (see Fig. 4). As described in [8], DICY exhibits an asymmetric peak shape (see Fig. 4a). The binding

energy of 399.9 eV is similar for the amine and the nitrile group [8,39–41]. In contrast N in the imine group is associated with the N1s peak at a lower binding energy of 398.6 eV [8,39,41]. The theoretical ratio I_{398.6}:I_{399.9} ((amine+nitrile)/imine) is 0.33 for unreacted DICY. Previous investigations revealed a change in the I_{398.6}:I_{399.9} ratio and a new peak at 400.6 eV for protonated amine (NH₃⁺) for DICY in direct contact with Sn, Al and Zn metals [8]. The increase of intensity at 398.6 eV was due to a subpeak at similar binding energy as imine, which was presumably related Lewis bonding of metal ions to DICY and was labeled as α in Fig. 4.

As depicted in Fig. 4b, the strongest increase of the side peak attributed to protonated amine (NH₃⁺) or hydrogen bonds at 400.6 eV [8,13,41,42] was ascertained for Sn. The side peak was marked as NH₃⁺ in Fig. 4. Moreover, a minor increase of the I_{398.6}:I_{399.9} ratio was observed. Hence, imine groups were formed on the surface of the Sn sample, most likely due to Lewis bonds of metal cations to the amine group, associated with the formation of an imine group [8,13–15]. This side peak at 398.6 eV was marked with imine/ α in Fig. 4.

For Ti, the I_{398.6}:I_{399.9} ratio did not change significantly. However, a further peak was emerging at a binding energy of 407 eV (see Fig. 4c) attributed to a nitro group [43–45]. Most likely, a corresponding compound is formed due to the catalytic properties of TiO₂. The formation of the nitro group was ascertained just for Ti, but not for Sn, Fe, Al, Zn or Mg.

The N1s peak of the Fe sample immersed in the H₂O + DICY mixture revealed changes which were in agreement with previous findings for Sn, Al and Zn [8]. A weak side peak attributable to protonated amine or hydrogen bonds was ascertained at 400.6 eV as well. Furthermore, the I_{398.6}:I_{399.9} ratio was significantly increasing (see Fig. 4d).

For Al and Zn, the increase of the imine/ α peak was even more noticeable (see Fig. 4e and f). However, no or only a rather weak protonated amine peak at 400.6 was obtained for Al and Zn immersed in the boiling H₂O + DICY solution.

Finally, changes of the N1s peak were also detectable for the immersed Mg sample. The protonated amine (NH₃⁺) peak was rather weak. In contrast the imine/ α side peak got very strong for Mg and even exceeded the main amine/nitrile peak at 399.9 eV (see Fig. 4g).

The results confirmed clearly that the interaction of aqueous DICY is dependent on the IEP of the metal oxides and hydroxides. Hence, the relationship between the IEP of various metal oxides and hydroxides, the NH₃⁺ and the imine/ α versus the amine/nitrile groups is illustrated in Fig. 5. At higher IEP of the metal oxide or hydroxide, the intensity of the NH₃⁺ peak at 400.6 eV is less pronounced or not discernible. Vice versa, a tendency for less imine/ α species (at 398.6 eV) was observed at lower IEPs of metal oxides or hydroxides. Due to the fact that nitro compounds were just formed for the Ti metal, Ti was not following this trend. Ascribed to significant scatter of the peak ratios and the IEP values, no linear or exponential fits were applied.

To reproduce the observed trends, the immersion tests and XPS analyses were repeated. Slight deviations of the N1s spectra were detected, indicated by the vertical error bar. The upper and lower bounds of the bar represent the measured data points, while the symbol is describing the averaged value. The data for Sn, Al and Zn were taken from a previous work [8] and reproduced. The lowest amount of N on the surface of the immersed metal sample was obtained for Al (<0.6 at.% and 1.5 at.%). Hence, the Al outlier in Fig. 5a is presumably related to measurement and fitting peak errors.

3.2. Chemical structure of epoxy coatings on metal substrates

Cryo-ULAM tapers as schematically depicted in Fig. 6 were prepared and characterized by XPS to assess the chemical structure of the surface, the cross-section of the epoxy coating and the interface of the coatings and the metal substrates. As the coating is only a few μ m thin (< 10 μ m) and it is well known that metal organic reaction products are highly mobile, the epoxy is considered in this study as an interphase and not as

Table 1
Isoelectric point (IEP) of relevant metal oxides and hydroxides [25].

Metal oxide or hydroxide	IEP [25]
SnO ₂	4.1–4.5
Sn(OH) ₄	4.0–5.2
TiO ₂	4.5–7.3
Fe ₂ O ₃ (hematite)	6.9–10
FeOOH (goethite)	6.9–9.4
Al ₂ O ₃	8.5–9.6
AlOOH	8.6–9.5
ZnO	8.0–9.8
Zn(OH) ₂	9.6–10.0
MgO	10.0–12.5
Mg(OH) ₂	10.8–12.0

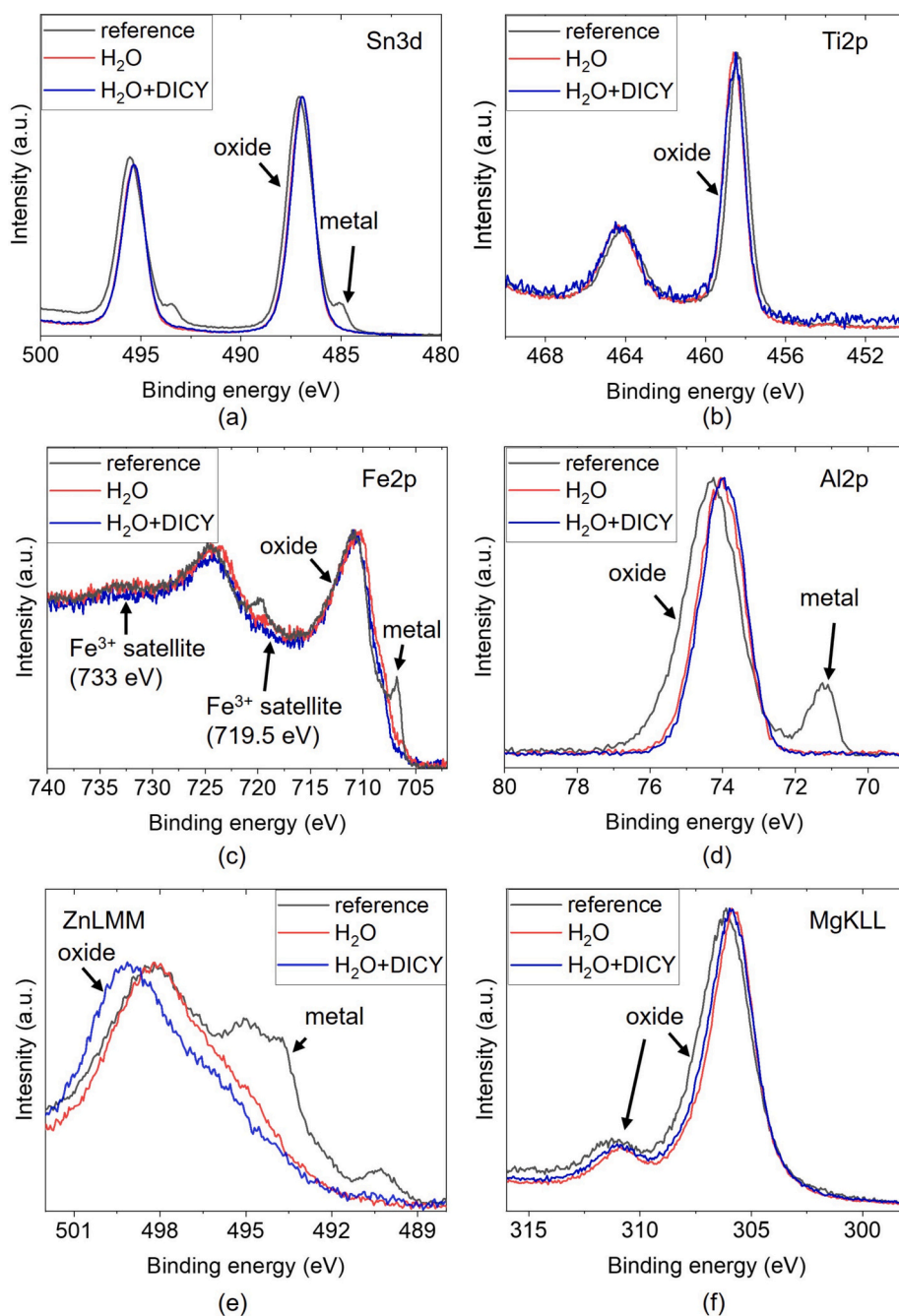


Fig. 3. High resolution XPS spectra of the metallic photoelectric and Auger peaks in the reference state and after immersion in boiling water or the H_2O + DICY mixture of (a) Sn, (b) Ti, (c) Fe, (d) Al, (e) Zn and (f) Mg.

a pure epoxy layer. The length of the cuts was several hundred μm . Due to the waviness of the coating and the slightly different angles of the sample holder, the cut lengths varied between the specimens. Linescans from the surface via the interphase to the epoxy/metal interface are depicted in Fig. 6 for Sn, Ti, Fe, Al, Zn and Mg substrates. The atomic concentrations of carbon, oxygen, nitrogen and the respective metals (Sn, Ti, Fe, Al, Zn and Mg) are plotted. In agreement with previous findings for epoxy coated Sn, Al and Zn sheets, a slightly lower nitrogen content was detected for Fe at the surface of the epoxy coating compared to the interphase [8]. In contrast, the N signature was comparable for the surface and the interphase of the epoxy on Ti and Mg. Interestingly, a similar N content was detected in the interphase independent on the metal substrate. The slight deviations were within the XPS accuracy limits. At the uncovered substrates, labeled as epoxy/metal interface

(see Fig. 6), the amount of C was still rather high (50 to 75 at.%). As discussed in [8], the original surface roughness of the metal substrates as well as grooves induced by the glass knives, resulted in thin epoxy layers remaining on the metal substrate, which revealed information of areas close to the epoxy/metal interface.

For the cut of the Sn sample, a higher N content was detected near the epoxy/Sn interface (see Fig. 6a). Hence, the epoxy coated Sn sample revealed an accumulation and enrichment of the DICY hardener at the interface. Furthermore, traces of Sn were discernible at the surface of the coating and in the interphase.

For coated Ti, the concentration of N on the surface was similar on the surface and in the interphase (see Fig. 6b). Near the interface, the N content was slightly increasing for the Ti substrate, related also to TiN species. The TiN species was confirmed in the corresponding high

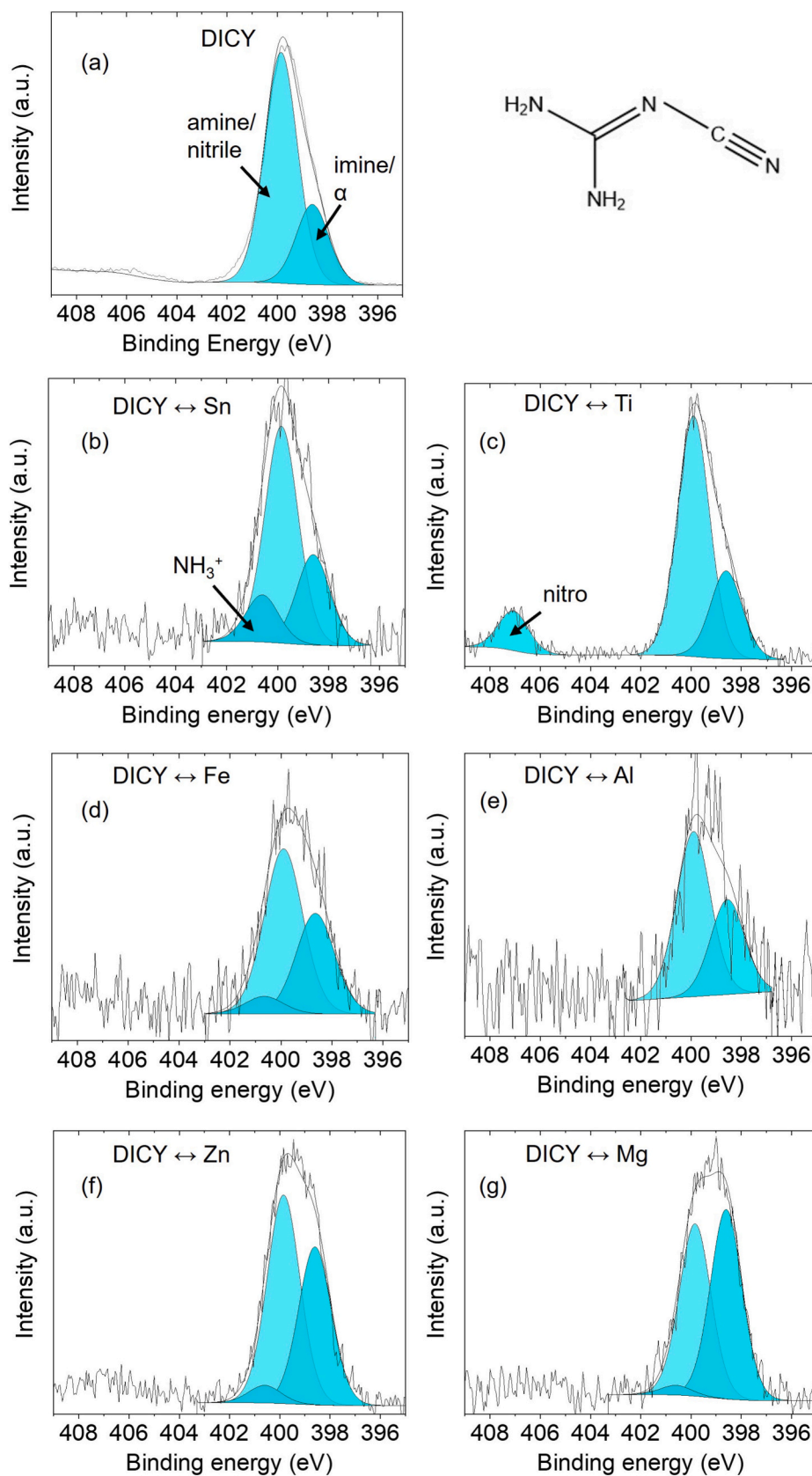


Fig. 4. N1s spectra of (a) DICY as well as (b) Sn, (c) Ti, (d) Fe, (e) Al, (f) Zn and (g) Mg immersed in the boiling H_2O + DICY solution.

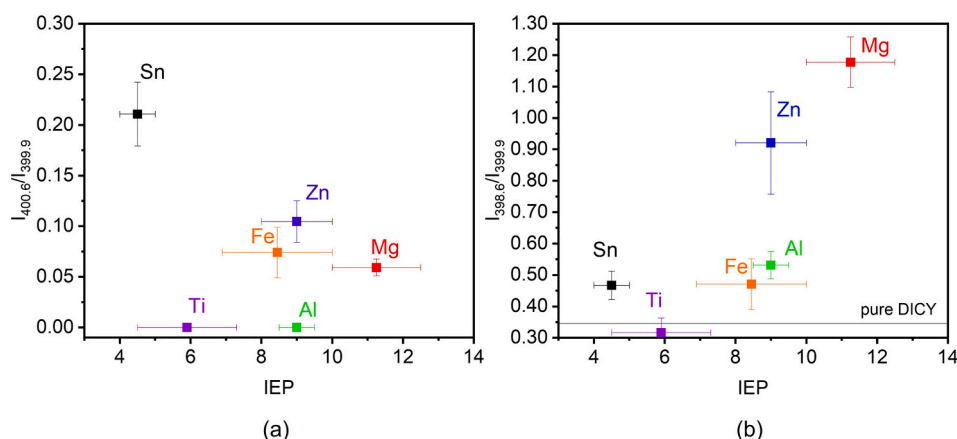


Fig. 5. Correlation between (a) protonated amine NH_3^+ to amine/nitrile ratio ($I_{400.6}/I_{399.9}$) and (b) the imine/ α to amine/nitrile ratio ($I_{398.6}/I_{399.9}$) and the isoelectric point (IEP) of oxides and hydroxides of the investigated metals Sn, Ti, Fe, Al, Zn and Mg.

resolution N1s spectrum taken near the interface (not shown). Strong side peaks at 407 and 396.5 eV were observed, which were attributable to nitro species and TiN, respectively. Interestingly, the TiN peak was not detected for the Ti samples immersed in an aqueous DICY solution (Fig. 4c). Hence, the formation was favored by further ingredients of the epoxy varnish (e.g., epoxy, emulsifier) or by damage of the Ti oxide passivation layer during ULAM cutting with the glass knife. Furthermore, low amounts of Na and Ca residues were detected in the interphase, but also near the epoxy/Ti interface. Interestingly, <1 at.% of Ti was also detected in the interphase.

The epoxy coated Fe sample revealed a lower N content at the surface (see Fig. 6c). In the interphase, the N content was in agreement with the other coated specimens about 1 at.%. At the interface, the amount of Fe was increasing and levelling off. Moreover, S and Si residues or contamination were discernible close to the interface. Due to a constant N concentration within the interphase but also near the interface, no DICY hardener was accumulating or enriching at the epoxy/Fe interface.

Compared to Sn, Ti or Fe, the layer of the Al coated sample revealed a higher content (~1at%) of metal in the interphase (see Fig. 6d). This is a clear indication for dissolution of Al from the substrate and migration into the epoxy coating. N was detected in the interphase, but not on the surface of the coating. Near the epoxy/Al interface the concentration of N decreased slightly. Thus, no enrichment of DICY was ascertained at the interface.

Similar to Al, also the epoxy coated Zn sample showed a relatively high amount of metal on the surface and within the interphase (see Fig. 6e). Again, no clear indications were deduced for an accumulation of N or DICY at the interface.

Finally, for the epoxy coated Mg sample, a rather high amount of Mg was observed on the surface and within the interphase of the epoxy coating (see Fig. 6f). Similar to Zn and Al, dissolution of metal ions and migration to the bulk resulting in the formation of an interphase or onto the surface of the epoxy coating is hypothesized. The higher the IEP of the metal oxide, the more pronounced was this effect. Near the interface of Mg, no accumulation of DICY was deduced.

In the interphase the amine/nitrile peak at 399.9 eV was dominant for all coated metal sheets. This is in agreement with previous results for Sn, Al and Zn [8]. A higher amount of imine/ α (398.6 eV) species in the interphase was observed for the coated Mg sample. This is a further indication for the formation of Mg-DICY complexes diffusing into deeper regions of the epoxy coating.

3.3. Crack growth kinetics and failure mechanisms of epoxy/metal laminates

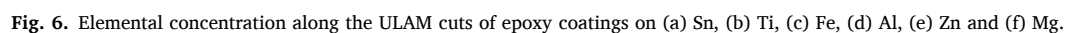
Fatigue crack growth curves of the epoxy/metal laminates based on

pure Ti, Fe, Al, Zn and Mg substrates are depicted in Fig. 7a. The low yield strength of the Sn substrate did not allow for fatigue crack growth testing. Pronounced plastic deformation of the Sn adherend was occurring. Consequently, the energy dissipated into the Sn substrate could not clearly be distinguished from the potential energy released due to crack propagation. Nevertheless, the other metal substrates exhibited sufficient yield strength.

Linear elastic fracture mechanics (LEFM) methods are applicable when the plastic zone ahead of the crack tip remains smaller than the specimen dimensions, such as crack length and adhesive thickness. The plastic zone is dependent on the applied strain energy release rate as well as the Young's modulus and the yield strength of the adhesive [46]. Assuming a conservative yield strength of 50 MPa for the epoxy varnish, a plastic zone diameter of around 60 μm was calculated at the beginning of the test at a strain energy release rate of 200 J/m^2 . In contrast, the plastic zone diameter was reduced to approximately 10 μm in the threshold region. Since the plastic zone remained significantly smaller than the crack length and was also smaller than the adhesive thickness at the fatigue threshold region, the validity of using LEFM approaches was confirmed. Moreover, the reduced plastic zone size in the threshold region supports the use of the fatigue threshold value rather than the critical strain energy release rate in fatigue analysis.

The Ti, Fe, Al, Zn and Mg substrates revealed just elastic deformation, allowing for evaluation considering a linear elastic fracture mechanics approach. The crack growth experiments were reproduced. Except for the Al laminate, a good agreement was confirmed. The deviation of the Al laminate was presumably related to a poor wetting behavior associated with inhomogeneous distribution of the epoxy layer. For establishment of correlations between the threshold strain energy release rate value and the isoelectric point of the metal oxides or hydroxides, the better performing Al laminate was considered.

Significant differences in crack growth kinetics were detected for the investigated laminates. A higher resistance for delamination crack growth was ascertained for the laminates based on Fe, Al, Zn and Mg. In contrast, a poor delamination resistance was obtained for the Ti laminate. Presumably, nitro compounds at the interface had a negative impact on the fatigue crack growth kinetics. To establish correlations with the isoelectric point of the metal oxides and hydroxides, the threshold strain energy release rate (G_{th}) was evaluated by Hartmann-Schivy fitting [34,47] of the crack growth kinetics data. A good correlation between threshold G values and the IEP of the metal substrates was obtained (see Fig. 7b). The higher the IEP of the metal oxide or hydroxide, the higher was the deduced G_{th} value. Especially, laminates based on Ti (IEP: 4.5–7.3) and Mg (IEP: 10.0–12.5) exhibited differences in the G_{th} value by a factor of 8 (5 vs. 42 J/m^2). Hence, Mg laminates are characterized by a lower tendency for crack initiation. For laminates



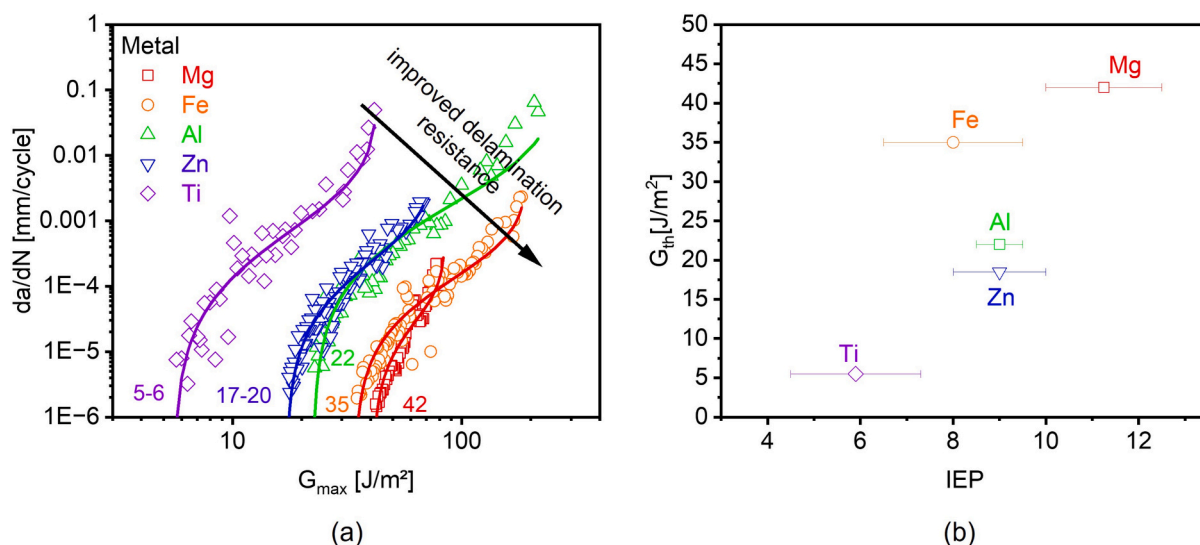


Fig. 7. (a) Fatigue crack growth curves of epoxy/metal laminates and (b) relationship between the threshold strain energy release rate (G_{th}) and the isoelectric point (IEP) of the metal oxides or hydroxides.

based on a more neutral IEP of 7 to 9, the correlation was not unambiguous. The poor correlation of the Fe, Al and Zn laminates were related to the wide range of IEP values stated in literature and the unknown exact IEP values for the investigated substrates. Nevertheless, a clear effect of the IEP on the fatigue crack growth characteristics was confirmed. The Mg laminate characterized by the highest IEP of the investigated metal substrates exhibited the highest G_{th} value, which was a factor of about 8 higher compared to the G_{th} of the lowest IEP Ti laminate [21].

Interestingly, a higher slope within the stable crack growth regime was obtained for the Mg laminate. By displacement-controlled testing data were gathered in a narrow strain energy release rate range. Most likely, this is an indicator for more brittle fracture of the Mg based laminates. The brittleness was presumably related to increased amount of metal complexes within the epoxy coating on Mg [48,49].

Visual inspection of the fractured surfaces confirmed a mixed cohesive and interfacial failure for the investigated laminates (see Fig. 8). Furthermore, the position of the failure was alternating between the two epoxy/metal interfaces. Hence, the crack jumped in and perpendicular to the crack growth direction from one interface to the other. Since the structures on the fractured surfaces is different, the number of jumps between the two interfaces was different for the laminates.

The fractured surfaces were also investigated by XPS and revealed similar spectra. Unfortunately, the interfacial and cohesively failed domains could not be distinguished by the camera system of the XPS device. Furthermore, the size of the domains was smaller than the XPS spot size of 400 μ m. So, in most of the investigated areas a mixed failure was identified by distinct metal peaks in the XPS spectra (see Fig. 9).

For the fractured Ti laminate low concentrations of Ti (1.7 at.%)

were discernible (see Fig. 9a), originating from partly interfacial failure at the metal surface and the epoxy adhesive. A peak for C—C at 285.0 eV and a peak at 286.7 eV for C—O were detected in the C1s spectra. The C—O peak was noticeably increased compared to the epoxy pre-polymer. Presumably, the crack was propagating through polyether based emulsifier domains, since the Poloxamer based emulsifier showed a high concentration of C—O bands. In detail a C-O/C-C ratio of 0.94 was obtained for the fractured surface of the Ti laminate. In comparison, the C-O/C-C ratio of the epoxy pre-polymer was just 0.49. Furthermore, a new peak was observed at 288.7 eV which was attributable to an ester (O-C=O) group. From the N1s spectra a ratio $I_{398.6}/I_{399.9}$ of 0.36 was obtained, which is comparable to the DICY reference. Interestingly, no nitro group was detected. Presumably, the concentration of DICY reacted at the interface was below the detection limit.

A higher concentration of metal was found for the fractured Fe laminate (see Fig. 9b) with a concentration of 6.5 at.%. The C-C/C-O ratio was comparable to the epoxy pre-polymer. Relatively high concentrations of O-C=O groups were obtained. Furthermore, the concentration of N was increased (7.7 at.%). The ratio of $I_{398.6}/I_{399.9}$ was 0.69, which is related to the strong imine/ α peak. In agreement with results of immersed substrates and the ULAM cut of coated Fe, a strong imine/ α peak was confirmed on the fractured surface. No protonated amine was detectable.

The surface of the fractured Al laminate revealed an Al content of 3.0 at.% (see Fig. 9c). Furthermore, a high amount of C—O bonds was deduced, indicating again a crack propagating through emulsifier rich domains. The ester peak was rather weak. Besides, the ratio $I_{398.6}/I_{399.9}$ of the N1s peak was 0.38 which was comparable to the one of the fractured Ti surface.

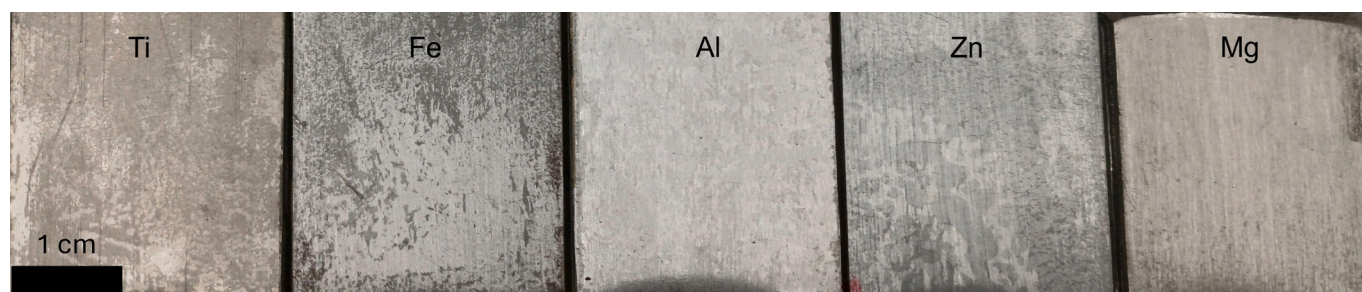


Fig. 8. Photographic images of the fractured surfaces, the threshold region is at the bottom of the image.

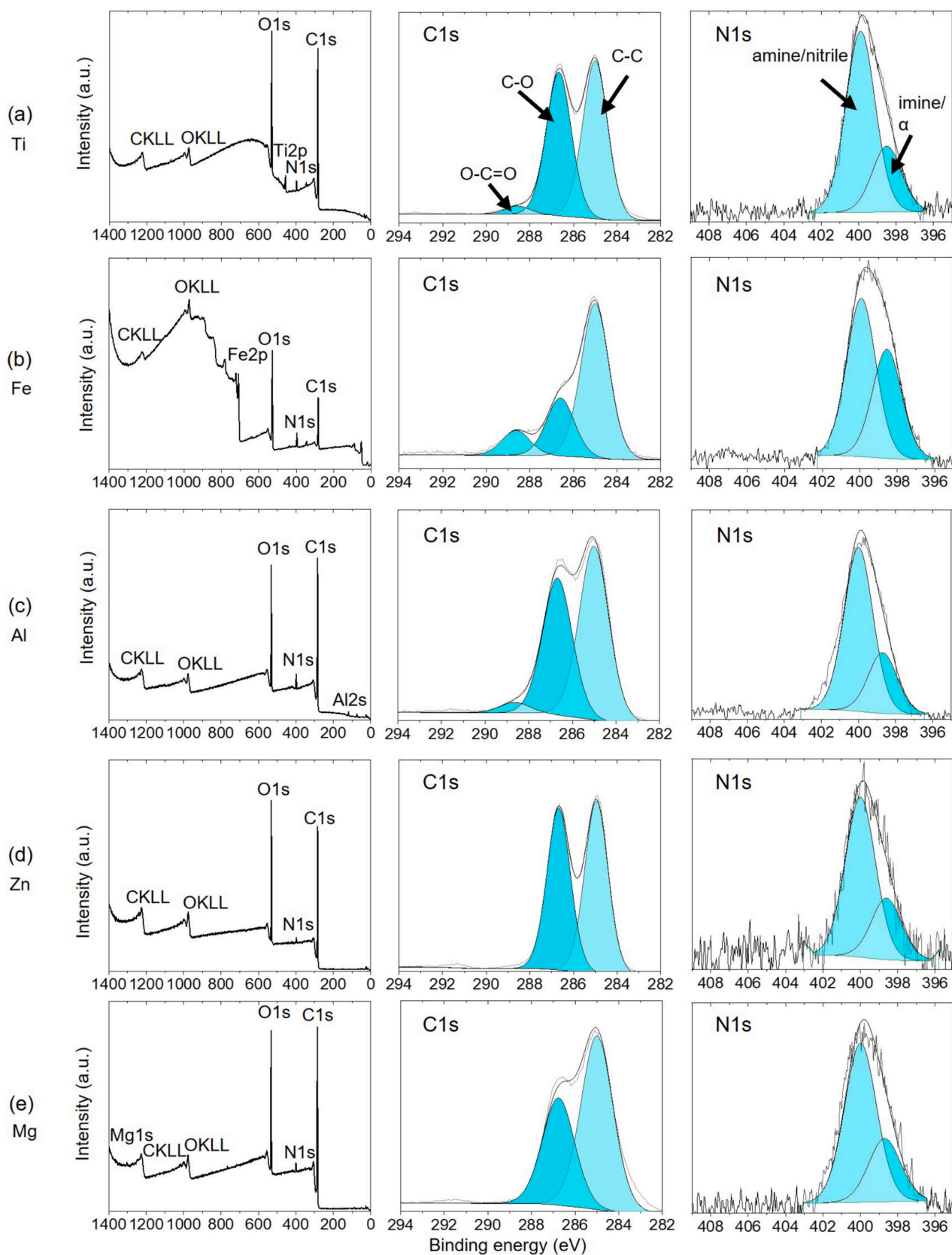


Fig. 9. XPS survey, as well as C1s and N1s spectra of fractured surfaces of (a) Ti, (b) Fe, (c) Al, (d) Zn and (e) Mg laminates.

No Zn traces were found on the fractured surface of the Zn laminate. This is a clear indicator for cohesive failure within the epoxy ply. Again, the content of C—O groups was rather high. Hence, preferred fracture within the emulsifier phase was detected. The N1s spectra were comparable to the DICY reference with an $I_{398.6}/I_{399.9}$ ratio of 0.37.

Finally, the fractured surface of the Mg laminate was characterized by a rather weak Mg1s peak (0.6 at.%)

(see Fig. 9e). Also here, crack propagation through emulsifier rich domains was assumed due to a high C—O/C—C ratio of 0.62. Again, the N1s peak was indicative for unreacted DICY. Interestingly, no clear signatures of Mg-DICY complexes were discernible on the fractured surface of the Mg laminate. Hence, a Mg complex rich phase within the epoxy layer was not uncovered by the propagating crack.

4. Summary and conclusions

In this work the interactions of different metals with aqueous DICY, epoxy coatings and their effect on the fatigue crack growth kinetics of laminates was investigated. While there are many different parameters affecting interfacial reactions, in this study special focus was given to the isoelectric point (IEP) of the substrate. The significance of this parameter was confirmed. On a fundamental level, the interaction of the DICY curing agent and the metal substrates of different IEP were examined by immersion of the metal sheets in a $H_2O + DICY$ solution. While protonated amine or hydrogen bonds were confirmed for low IEP substrates, high IEP substrates were characterized by imine groups or covalent bonds of DICY to metal ions.

Since the metal oxide was in contact with an aqueous solution, metal ions were presumably dissolved into the aqueous liquid associated with the formation of organic complexes [7,8,11–15]. In the case of low IEP substrates, the protonated amines at the interface were presumably acting as a diffusion barrier for metal ions. However, a high IEP resulted in a positive surface charge and negative ions near the surface, which presumably facilitated the diffusion of metal ions (see Fig. 10). Investigations on coated metal substrate level by cryo-ULAM and XPS confirmed this mechanism deduced from immersion tests. For coated metals with a high IEP point noticeable concentrations of the metal were detected on the surface and within the interphase of the epoxy coating. Moreover, an enrichment of N and hence, protonated amine, was ascertained especially for the coated Sn substrate with an acidic IEP.

Investigations on the laminate level confirmed a direct impact of the epoxy/DICY/metal interactions on the fatigue growth kinetics. A higher IEP of the substrate led to a better delamination resistance and enhanced strain energy release rate values in the threshold regime (see Fig. 10). Due to less accumulation of low molar mass DICY molecules at the interface, the crack was preferentially propagating in a cohesive manner within the epoxy ply. Interestingly, the crack was mostly propagating within the emulsifier rich phase, not uncovering the metalorganic complexes within the epoxy. The lower the IEP of the metal substrate, the more interfacial failure was discernible associated with significant metal residues on the fractured surfaces. Nevertheless, the crack was also going through the epoxy layer and alternatingly propagating at the weaker interface. The fractured surfaces of Ti and Fe laminates were more inhomogeneous, even on a macroscopic photographic level.

In the future, the fracture patterns could be elucidated in more detail also by confocal light or scanning electron microscopy. Moreover, it would be of interest to perform an even more comprehensive study considering further pure metals such as copper or metal alloys. Finally, it is emphasized that meanwhile established environmental fracture mechanics could also be employed to examine local ageing effects under hot-humid conditions or in water-based fluids differing in the pH value.

CRediT authorship contribution statement

Gary Säckl: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Gabriel Riedl:**

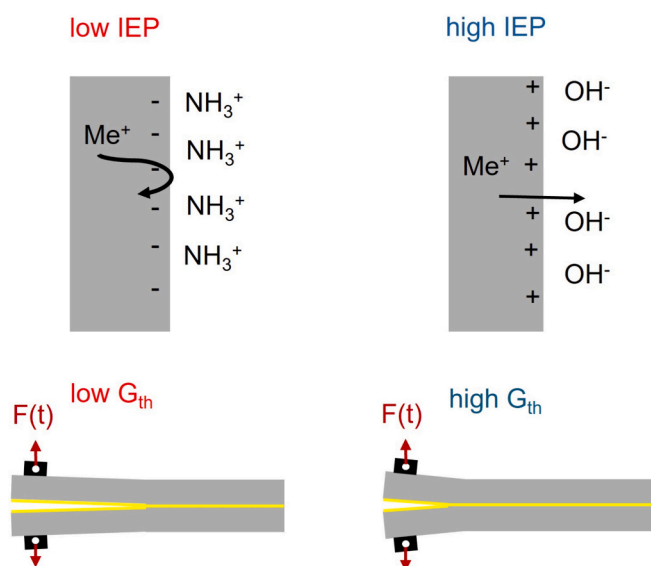


Fig. 10. Schematic sketch of metal surfaces with low (left) or high (right) IEP in contact with $H_2O + DICY$ mixtures or waterborne epoxy varnishes with DICY hardeners and the effect of the IEP on the delamination resistance.

Writing – review & editing, Methodology, Investigation. **Jiri Duchoslav:** Validation, Supervision, Formal analysis. **Cornelia Marchfelder:** Resources, Methodology. **Maëlen Aufray:** Writing – review & editing, Conceptualization. **David Stifter:** Writing – review & editing, Validation, Supervision. **Gernot M. Wallner:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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